

Accelerated Screening of Halide Double Perovskites via Hybrid Density Functional Theory and Machine Learning for Thermoelectric Energy Conversion

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A comprehensive first-principles and machine learning study is conducted on 102 halide double perovskites to identify promising candidates for thermoelectric applications. The HSE06 hybrid functional within the Quantum ATK framework is used to accurately determine electronic structures, bandgaps, and total and partial densities of states. Boltzmann transport theory is applied to figure out important thermoelectric parameters, such as the Seebeck coefficient, electrical conductivity, and ZT values over a wide range of temperatures. Supervised machine learning models are trained on density functional theory (DFT)-derived descriptors to speed up the discovery of new materials. These models demonstrate high predictive accuracy for thermoelectric performance across different chemical spaces. A detailed analysis of the electronic band structures and orbital contributions is carried out for Rb_2GeI_6 , Rb_2PbI_6 , Cs_2SnBr_6 , and In_2PtCl_6 , some of the best-performing compounds. A wide range of behaviors is observed, including metallic, degenerate, and wide-bandgap semiconducting, which correlate with distinct transport properties. This unified method shows how using accurate DFT, transport theory, and machine learning together can help find new materials with specific functions. This will lead to the development of next-generation thermoelectric technologies based on environmentally friendly halide perovskites.

thermoelectric material is quantified by its dimensionless figure of merit, $ZT = \frac{S^2\sigma T}{\kappa}$, where S is the Seebeck coefficient, σ the electrical conductivity, T the absolute temperature, and κ the thermal conductivity. To achieve the optimal thermoelectric performance, we need to maximize the power factor $S^2\sigma$ and minimize thermal conductivity. This is often hard to do because these two transport parameters depend on each other. In the past, finding and improving thermoelectric materials relied on trial and error in the lab and theoretical modeling, especially using density functional theory (DFT) with Boltzmann transport equations. These methods have produced high-performance compounds like Bi_2Te_3 , PbTe , SnSe , and various half-Heusler and skutterudite materials. However, their computational cost becomes excessive when investigating complex chemistries, extensive compositional spaces, or heavily doped and disordered systems. This has motivated a growing interest in machine learning (ML) approaches, which offer the ability to rapidly screen vast materials spaces, predict

1. Introduction

Thermoelectric materials hold significant promise in addressing the global challenge of energy efficiency by enabling direct conversion between thermal and electrical energy. These materials are especially valuable for waste-heat recovery, space power systems, and solid-state refrigeration. The performance of a

key transport properties, and extract underlying structure–property relationships that are often inaccessible through conventional approaches.^[1–6]

In recent years, there has been an abundance of research that combines ML with the discovery and design of thermoelectric materials. ML models have shown great accuracy in predicting Seebeck coefficients, electrical and thermal conductivities, and

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even the compound figure of merit ZT itself by using curated datasets made up of either experimental measurements or high-throughput DFT outputs. Song et al. put together a dataset of more than 5,000 experimental measurements for 880 compounds, which is a well-known example. The trained models they used got coefficients of determination higher than 0.9 for all thermoelectric properties and a mean absolute error lower than 0.06 for ZT. This shows that data-driven models can capture the complex nonlinearities that are common in thermoelectric behavior.^[7]

There are many supervised learning algorithms that have been tested for predicting thermoelectric properties. These include support vector machines, Gaussian process regression, random forests, and deep learning models. Deep learning architectures based on composition, like those that employ attention mechanisms or graph-based encodings, have been able to predict transport behavior using only elemental information. For instance, Antunes et al. created a deep neural network model that only used stoichiometric input to predict the Seebeck coefficient, conductivity, and power factor at different temperatures and doping levels. They were able to do this with a high level of accuracy while testing new binary and ternary selenide compounds.^[8] Other research has demonstrated that decision-tree ensembles, like random forests, frequently surpass deep networks in small to medium-sized datasets, offering the additional benefit of interpretability. These models have independently rediscovered established high-performing materials such as NbFeSb and suggested new candidates like TiGePt and ZrInAu, which were subsequently confirmed through first-principles calculations.^[9] ML not only helps predict properties, but it also speeds up discovery through high-throughput screening pipelines. These usually involve training models on data that already exists and then using them to look at compositions that have not been studied yet or that are just ideas. In multiple instances, ML models have detected atypical chemical families exhibiting elevated predicted ZT, subsequently validated by experiments or DFT.^[10] The Citrine Thermoelectrics Engine, created in 2016, is a good example of this ability because it predicted that a group of rare-earth cobalt bismuth compounds would have good thermoelectric performance, which was later shown to be true in lab tests.^[11] In addition, ML makes it easy to prioritize candidates in the materials genome, which cuts down on the cost of computation compared to exhaustive DFT screening.

But ML is useful for more than just making predictions and screening. Researchers have learnt more about the physicochemical factors that affect thermoelectric performance by looking at model outputs like feature importance or SHAP values. For example, atomic mass, carrier concentration, electronegativity variance, and bonding topology are some of the most important factors that can predict lattice thermal conductivity or Seebeck behavior. In contemporary categories of thermoelectric materials, such as ionic thermoelectrics and disordered polymer systems, ML has facilitated the discovery of previously unrecognized correlations between molecular characteristics and thermopower. Wu et al. employed neural models trained on SMILES descriptors to illustrate that molecular flexibility and hydrophobicity adversely affect Seebeck coefficients, a conclusion supported by molecular dynamics simulations.^[12]

Nonetheless, numerous obstacles persist in the application of ML to thermoelectric materials. Some of these are the quality and lack of data, especially for new material systems; the fact that it is hard to apply what we do know to new chemistries; and the fact that it is hard to balance model complexity and interpretability. Advanced models like graph neural networks can capture subtle structural dependencies, but they usually need large training sets and well-defined crystal structures. Also, predictions based only on data may not be physically consistent or may create artifacts when used outside of their training domain. To deal with these problems, hybrid methods that combine ML with physics-based modeling, like Boltzmann transport solvers or phonon calculations, are becoming more popular. Transfer learning, active sampling, and domain-informed featurization are some of the methods that are becoming more common to make models more general and robust.

Recently, lead-free double perovskites of the form A_2BX_6 have shown promise as thermoelectric materials because they have low lattice thermal conductivity and electronic structures that can be changed to give large Seebeck coefficients.^[13] First-principles (DFT) calculations on halide A_2BX_6 compounds (e.g., Cs_2SnI_6 , Rb_2SnI_6 , Rb_2PdI_6) demonstrate narrow-bandgap semiconducting characteristics with degenerate band extrema and low effective carrier masses—attributes that facilitate high power factors and improved thermoelectric performance.^[14] In fact, p-type doping in these perovskites is expected to produce Seebeck coefficients between 0.5 and 0.8 mV K^{-1} and a thermoelectric figure of merit ZT close to one when conditions are perfect.^[14] In addition to these first-principles insights, ML techniques have been used to speed up the discovery of new materials. For example, data-driven models trained on DFT results can quickly screen large compositional spaces and predict important properties (like formation energies and bandgaps) across the A_2BX_6 family.^[15–18] This combined DFT–ML approach is pushing the field forward by helping scientists find new high-performance A_2BX_6 perovskites and directing them toward specific experiments to test them. Consequently, these materials are progressively regarded as promising candidates for next-generation thermoelectric applications, including the utilization of industrial waste heat and the powering of wearable electronics and other ambient energy-harvesting devices.^[16]

In this article, we seek to identify and comprehend high-performance thermoelectric candidates from an extensive collection of vacancy-ordered double perovskites by integrating hybrid DFT calculations with Boltzmann transport theory and ML analysis. We use the HSE06 functional to get the electronic structures of 102 compounds right. Then we use Boltzmann transport modeling to look at their thermoelectric properties, which depend on temperature, such as the Seebeck coefficient, electrical conductivity, and figure of merit (ZT). We create predictive ML models that can find trends across chemical space and figure out how structure and property are related in order to speed up discovery and make it easier to understand. The study concludes by emphasizing the electronic properties of high-performing compounds and offering a data-driven framework for the targeted design of thermoelectric materials.

2. Experimental Section

2.1. DFT and Boltzmann Theory Computational Details

We used the QuantumATK simulation package^[19] to do first-principles calculations on a dataset of 102 halide double perovskite compounds. By using DFT and semi-classical Boltzmann transport theory together, we were able to look at the structural, electronic, and transport properties. The Heyd–Scuseria–Ernzerhof (HSE06) hybrid functional^[20,21] was used to accurately describe the electronic structures. This functional is known to give more accurate bandgaps and dispersion characteristics than standard GGA functionals, especially for semiconductors and wide-bandgap materials. We used a linear combination of atomic orbitals basis set and a Monkhorst-Pack k-point grid (usually $9 \times 9 \times 9$) to optimize the structure. This made sure that the total energy and force converged within $0.01 \text{ eV } \text{\AA}^{-1}$.^[22] The band structures and density of states (DOS) data were exported from QuantumATK and used as input for the BoltzTraP code,^[23] which implements semi-classical Boltzmann transport theory under the constant relaxation time approximation (CRTA). This allowed us to compute temperature- and chemical potential-dependent transport properties.^[24,25]

In this framework, the Seebeck coefficient (S), electrical conductivity (σ), and electronic thermal conductivity (κ_e) were directly obtained from BoltzTraP calculations as functions of temperature and chemical potential. The dimensionless figure of merit was subsequently evaluated as $ZT = S^2 \sigma T / \kappa_e$ within the CRTA. A single constant relaxation time (τ) was assumed for all compounds and not explicitly calculated or fitted, which is a common practice in high-throughput thermoelectric screening. This ensures that the results provide reliable relative trends and rankings across different A_2BX_6 compounds, even though the absolute values of σ and κ_e scale with τ . Importantly, only the electronic thermal conductivity (κ_e) was considered in this study, while the lattice contribution (κ_l) was not included. Because κ_l is often significant in halide perovskites with soft phonon modes, the reported ZT values should be interpreted as upper-bound estimates. This simplification is consistent with CRTA-based screening approaches, where the aim is to capture electronic-structure-driven trends and identify promising candidates for further detailed investigation.

We constructed the datasets of vacancy-ordered double perovskites by considering A_2BX_6 -type compounds, where “X” represents a monovalent halide anion, “A” is a monovalent organic or inorganic cation, and “B” is a divalent transition metal cation. All selected compounds crystallize in the antiperovskite structure with the standard Fm-3m (No. 225) space group. As shown in **Figure 1**, the conventional cubic unit cell of a pristine perovskite places A atoms at the 8c ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}$) positions, B atoms at 4a (0, 0, 0), and halide anions at 24e ($x, 0, 0$) Wyckoff sites. In this arrangement, the cations on the A-site are 12-fold coordinated, and the atoms on the B and X sites are sixfold coordinated. This makes separate BX_6 octahedra with halide anions at the corners. The vacancy-ordered double perovskites also have a structure that is made up of BX_6 octahedra and empty B-sites that alternate with each other. In A_2BX_6 compounds, the $[BX_6]$ octahedra expand to make room for bigger ions, which makes the gaps between the octahedra smaller. The vacancy-ordered double perovskite

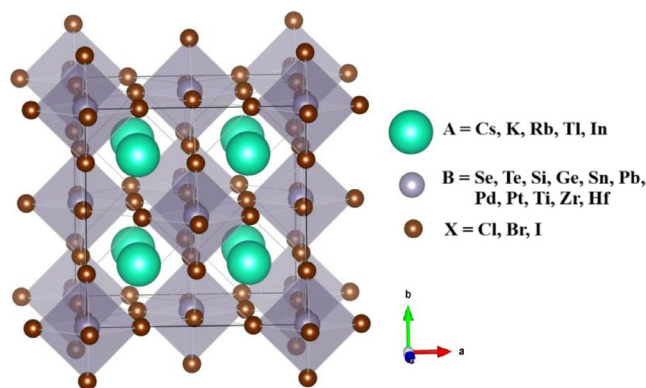


Figure 1. Crystal Structure of A_2BX_6 perovskite.

structure does not have B–X–B covalent bonding, but the close-packed anionic lattice still creates dispersive frontier electronic bands with light, mobile charge carriers.

Vacancy-ordered double perovskite structures were generated computationally by substituting selected A, B, and X elements into known A_2BX_6 prototype frameworks, rather than being sourced from existing databases or literature. No chemical or structural filtering criteria (for example, ionic radius or valence constraints) were applied during the generation of these structures. After generation, the resulting atomic configurations were not subjected to any additional preprocessing or structural cleaning; the raw substituted structures were directly used as inputs for the DFT optimization described above.

2.2. Machine Learning Framework

2.2.1. Dataset Construction

The dataset comprises computed thermoelectric properties for a family of halide-based compounds across varying temperatures (100 K, 300 K, and 800 K, support information) and chemical potentials (μ). The following features were extracted at each μ : 1) Seebeck coefficient S ($\mu\text{V/K}$); 2) Electrical conductivity (EC) σ (S/m); 3) Thermal conductivity (TC) κ ($\text{W/m}\cdot\text{K}$); 4) Chemical potential μ (eV).

The target property is the dimensionless thermoelectric figure of merit ZT , where T is the absolute temperature in Kelvin. Seebeck values were converted from $\mu\text{V/K}$ to V/K before calculating ZT . All datasets were pre-processed to remove missing or unphysical values (e.g., negative or NaN values).

It is important to notice here that because all structures were fully relaxed prior to property evaluation, intrinsic parameters such as lattice constants, atomic masses, and bandgaps are already implicitly reflected in the computed thermoelectric descriptors (S , σ , κ , and μ). So, it would be redundant to add these parameters as separate inputs. To keep things clear and lower the chance of overfitting, given the size of the dataset, we intentionally limited the input space to this small set of physically meaningful features. This choice makes sure that the ML models stay strong and focused on the main factors that affect thermoelectric performance. Even though the four features (S , σ , κ , μ) are linked through transport physics, correlation analysis showed

that there is no exact multicollinearity between them. Each variable captures a distinct contribution to thermoelectric behavior, and redundancy was not observed at levels that would destabilize model training. This limited and physically meaningful feature set, therefore, ensures both stability and interpretability of the ML models. Furthermore, because the transport coefficients were evaluated under the CRTA with only κ_e considered, the absolute magnitude of ZT values may be overestimated. However, all compounds were treated consistently, ensuring that relative performance trends remain reliable. As such, the ML models are trained on internally consistent data and are intended to capture comparative rankings of candidate materials. While inclusion of lattice thermal conductivity or material-specific relaxation times would refine the quantitative predictions, the qualitative ranking of high-performing candidates is expected to remain unchanged. A systematic sensitivity analysis of these assumptions is an important direction for future work.

2.3. Machine Learning Algorithms

2.3.1. Random Forest Regressor

Random Forest (RF) is an ensemble learning method that constructs multiple decision trees and averages their predictions for improved generalization.^[26] It was used to model the relationship: $ZT = f(\mu, S, \sigma, \kappa)$.

Key hyperparameters: 1) Number of trees: 100; 2) Splitting criterion: mean squared error (MSE); 3) Bootstrapping: enabled; 4) Random seed: 42 for reproducibility.

The RandomForestRegressor class from scikit-learn^[27] was used to make RF models. Grid search with fivefold cross-validation over the training data was used to find the best RF hyperparameters. We changed the number of trees, the depth of the trees, and the splitting criterion. The final configuration (100 trees, MSE criterion) was chosen because it gave us high accuracy with little variance and did not overfit.

2.3.2. Neural Network (NN)

A feedforward artificial neural network was trained to predict ZT from the same input features. The architecture was as follows: 1) Input layer: 4 neurons (μ , S , σ , κ); 2) Hidden layers: two layers with 32 and 16 ReLU neurons, respectively; 3) Output layer: 1 neuron (ZT) with linear activation.

2.3.3. Training Details

1) Loss function: MSE; 2) Optimizer: Adam; 3) Batch size: 8; 4) Early stopping: patience = 20 epochs; 4) Framework: TensorFlow/Keras.

Before training, StandardScaler standardized all of the features. We figured out the NN architecture and training settings by tweaking them over and over while keeping an eye on the validation. We tested the number of hidden layers, the number of neurons, the batch size, and the learning rate in a systematic way, and we used early stopping to stop overfitting. We chose the final model (two hidden layers with 32 and 16 neurons, ReLU activation, and Adam optimizer) because it always had

the most stable convergence and the lowest validation loss across many tests.

We chose the RF and NN models because they give us different points of view. RF is good for small datasets, does not overfit, and is easy to understand because it shows feature importance. NN, on the other hand, can find complex nonlinear relationships between descriptors and ZT. Using these models together makes sure that predictive performance is not limited by just one algorithm. This is in line with what is already known in the field of thermoelectric materials screening.

2.4. Model Evaluation

Models were trained using a 75:25 train/test split. The following metrics were used for regression performance:

- **Coefficient of determination (R^2):**

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (1)$$

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2} \quad (2)$$

Parity plots are used to visually compare predicted versus actual ZT values on the test set.

2.5. Clustering Analysis

To group materials based on their maximum thermoelectric performance and μ -dependence, unsupervised clustering was performed using K-Means.^[27] Input features were: 1) Maximum ZT over μ range; 2) Corresponding μ value at peak ZT.

These features were standardized before clustering, and principal component analysis (PCA) was applied for 2D visualization of clusters.

2.6. SHAP Analysis

To quantify the contribution of each input feature to the predicted ZT, SHAP (SHapley Additive exPlanations) values were computed using TreeExplainer for RF models.^[28] SHAP assigns an importance value ϕ_j to each feature x_j :

$$f(x) = \phi_0 + \sum_{j=1}^M \phi_j \quad (3)$$

where ϕ_0 is the base value (expected model output) and ϕ_j is the marginal contribution of feature j . SHAP beeswarm plots were used to visualize the feature influence across all predictions.

2.7. Classification via Confusion Matrix

To evaluate the ability of models to classify ZT into performance categories ("Low", "Medium", "High"), ZT values were binned as: 1) Low: $ZT < 0.2$; 2) Medium: $0.2 \leq ZT < 0.4$; 3) High: $ZT \geq 0.4$.

Confusion matrices were generated for both RF and NN predictions.^[29,30] The accuracy of class prediction provided insight into the model's robustness in distinguishing material classes for thermoelectric applications.

3. Results and Discussion

3.1. Clustering Analysis of Thermoelectric Materials at Multiple Temperatures

We used the K-Means algorithm^[26] to do unsupervised clustering on thermoelectric materials based on their transport properties. This helped us understand how they are distributed and grouped. The input features, Seebeck coefficient (S), electrical conductivity (σ), and thermal conductivity (κ), were normalized, with κ inverted to indicate the preferred state of low thermal conductivity. PCA was used to reduce the number of dimensions so that the material space could be seen in two dimensions and understood more easily. Clustering was independently applied to datasets at 100 K, 300 K, and 800 K. In each case, three clusters were identified, corresponding to distinct performance regimes. The number of clusters was set to $k = 3$ after testing values of k between 2 and 6. Quantitative evaluation using the silhouette score and elbow method indicated that $k = 3$ achieves the

best trade-off between cluster compactness and separation, with higher k values offering only marginal improvements. Furthermore, the three clusters align well with physically distinct thermoelectric regimes: average performers, anomalous outliers, and high-performance candidates. This combination of statistical validation and physical interpretability justifies the choice of $k = 3$ for all reported analyses. The clustering results are presented in Figure 2.

Across all temperatures, three main clusters were consistently observed: 1) **Cluster 0** represents the bulk of the materials and occupies the central region of PCA space. This cluster comprises compounds with moderate values for S , σ , and κ , suggesting stable yet average thermoelectric performance; 2) **Cluster 1** appears sparsely populated and typically includes materials with extreme or anomalous values. One consistent member across all temperatures is K_2PtCl_6 , which exhibits unusual placement in PCA space, potentially reflecting anomalous transport behavior or outlier status; 3) **Cluster 2** forms a distinct and coherent group, generally containing high-performance thermoelectric candidates with strong Seebeck response, favorable electrical conductivity, and suppressed thermal conductivity. This cluster is spatially separated in PCA coordinates, indicating clear feature-based differentiation.

At 100 K, material clustering shows minimal spread in PCA space, implying that thermoelectric behavior is relatively

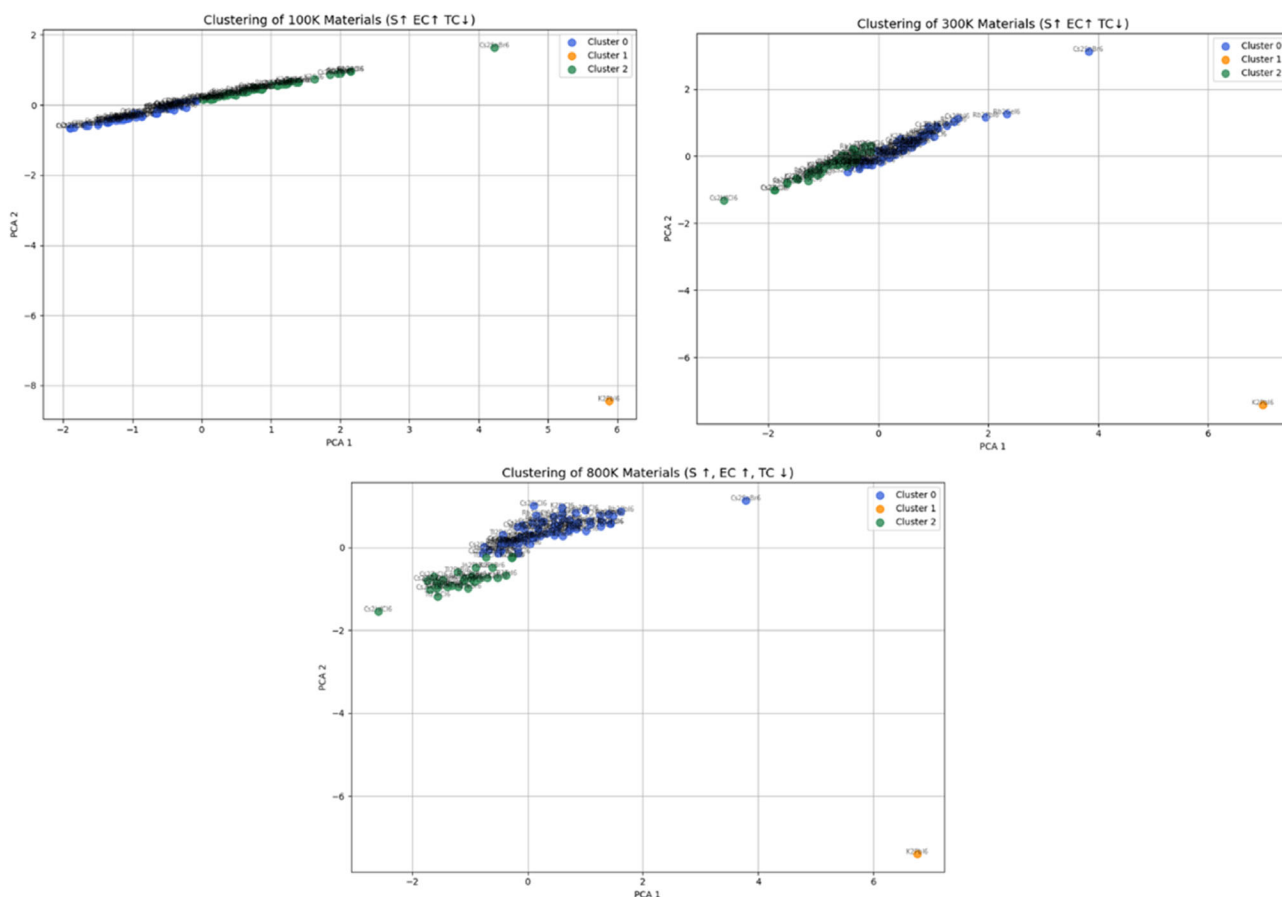


Figure 2. PCA clustering of 102 halide double perovskites based on thermoelectric features at 100 K, 300 K, and 800 K.

compressed at low temperatures. Materials exhibit small variations in electronic transport properties, and clustering reflects marginal feature separability.^[31,32] In contrast, at 300 K, clustering becomes more distinct. Cluster 2 expands to include materials such as Cs₂GeI₆ and Rb₂PbI₆, both of which were previously identified by supervised learning as high-ZT performers. This suggests enhanced feature-based separability and a growing distinction in thermoelectric behavior at mid-temperature. At 800 K, the PCA spread reaches its maximum. Phonon scattering and thermally activated carriers introduce higher variability in EC (σ) and TC (κ), enhancing clustering resolution. High-temperature clusters include robust materials, such as Rb₂GeI₆, Cs₂GeBr₆, and In₂PtCl₆, all demonstrating sustained high ZT across the full μ range.

The persistence of **Cluster 2** across all three temperatures suggests that these materials maintain their performance characteristics over a wide thermal range. Materials that consistently belong to this cluster are likely to exhibit both strong ZT and stability, making them compelling candidates for real-world thermoelectric applications. 1) **Cluster 2** consistently identifies high-ZT materials and matches ML model predictions; 2) **Cluster 1** likely contains materials with modeling uncertainty or unoptimized transport properties; 3) The movement of materials across clusters from 100 K to 800 K reflects their tunability and sensitivity to temperature, highlighting candidates for μ or carrier engineering.

These results (see **Table 1**) affirm the utility of unsupervised clustering in pre-selecting materials for focused design, and suggest that robust performers such as Rb₂GeI₆ and In₂PtCl₆ should be prioritized for further experimental or computational exploration.

3.2. Machine Learning Models for ZT Prediction

To evaluate the ability of ML models to classify thermoelectric performance, the computed figure of merit (ZT) was divided into three tiers: Low ($ZT < 0.2$), Medium ($0.2 \leq ZT < 0.4$), and High ($ZT \geq 0.4$). This categorization enables assessment of how effectively ML models distinguish between materials of varying practical relevance, particularly those suitable for thermoelectric applications.^[15,33–35] Confusion matrices were generated for both RF and NN models across three temperature regimes: 100 K, 300 K, and 800 K (**Figure 3**).

At 100 K, both models performed exceptionally well. The RF model correctly classified nearly all instances, with only a single medium-ZT material misclassified. All high-ZT and low-ZT cases were identified without error. The NN model mirrored this

high accuracy, perfectly classifying all high-ZT and low-ZT materials and misclassifying only one sample in the medium class.

These results demonstrate that the models are able to separate classes effectively, even at low temperatures where feature variation is relatively narrow. At 300 K, model performance remained strong. The RF model still correctly identified all high-ZT materials, but it got a little confused at the low-medium boundary. The NN model once more demonstrated marginally improved balance, accurately classifying nearly all high-ZT materials while preserving elevated overall accuracy. This observation corroborates the results of Kim et al.,^[36] who indicated that deep learning architectures exhibit robust performance across diverse thermal regimes in high-throughput thermoelectric screening tasks. The heightened challenge in distinguishing medium-ZT cases at this temperature is probably attributable to overlapping feature distributions, as the thermoelectric properties of numerous materials commence to evolve more dynamically at room temperature.^[30] At 800 K, the two models became more different from each other. The RF model was clearly biased toward the dominant low-ZT class. It correctly identified all low-ZT materials but did not recognize any high-ZT candidates. This shows that the model is sensitive to class imbalance, which is a known problem that can hurt performance on categories that are not well represented.^[30] The NN model, on the other hand, was able to correctly identify all high-ZT materials and did well with medium-ZT predictions, making only a few mistakes when it came to low-ZT samples. This shows how strong neural networks are at dealing with unbalanced datasets and learning subtle nonlinear feature relationships. Antunes et al.^[37] also noted this advantage when they used attention-based neural networks to predict thermoelectric transport properties from compositional features. A comparative summary of the classification results reveals that while both models are highly accurate under balanced conditions (e.g., at 100 K and 300 K), the neural network architecture is more resilient to data skew. At 800 K, where high-performing materials are less common, the NN model kept its sensitivity and recall, making it better for high-throughput screening in thermally difficult situations. We also looked at classification performance using precision, recall, and F1-score across the Low, Medium, and High ZT categories in addition to the confusion matrices. The Supplementary Information (Table S1, Supporting Information) shows that both models are very accurate overall. The RF model is better at getting the dominant classes right, while the NN model is better at getting the minority high-ZT category right. This complementary behavior shows how strong the dual-model approach is.

To evaluate the predictive efficacy of ML models in estimating the thermoelectric figure of merit (ZT), parity plots were created to compare predicted and actual ZT values for both RF and NN models at 100 K, 300 K, and 800 K (**Figure 4**). The coefficient of determination (R^2) and RMSE are shown on each plot to show how accurate the model is. At 100 K, both the RF and NN models agreed very well with the ground truth, with R^2 values of 0.951 and 0.974, respectively. The NN model had a slightly lower RMSE (0.0382) than the RF model (0.0523), which means that it was better at reducing errors. This performance is due to the fact that most materials show low phonon activity and little temperature-induced scattering at cryogenic temperatures, which means that their thermoelectric responses are not very wide. The uniformity

Table 1. Interpretation of clusters.

Cluster	Key Features	Physical Meaning
0	Balanced S, moderate σ , and κ	Average performers; thermally stable candidates
1	Outliers or extreme values	Materials with anomalous or poor behavior
2	High S, strong σ , low κ	Top thermoelectric candidates (ZT ↑)

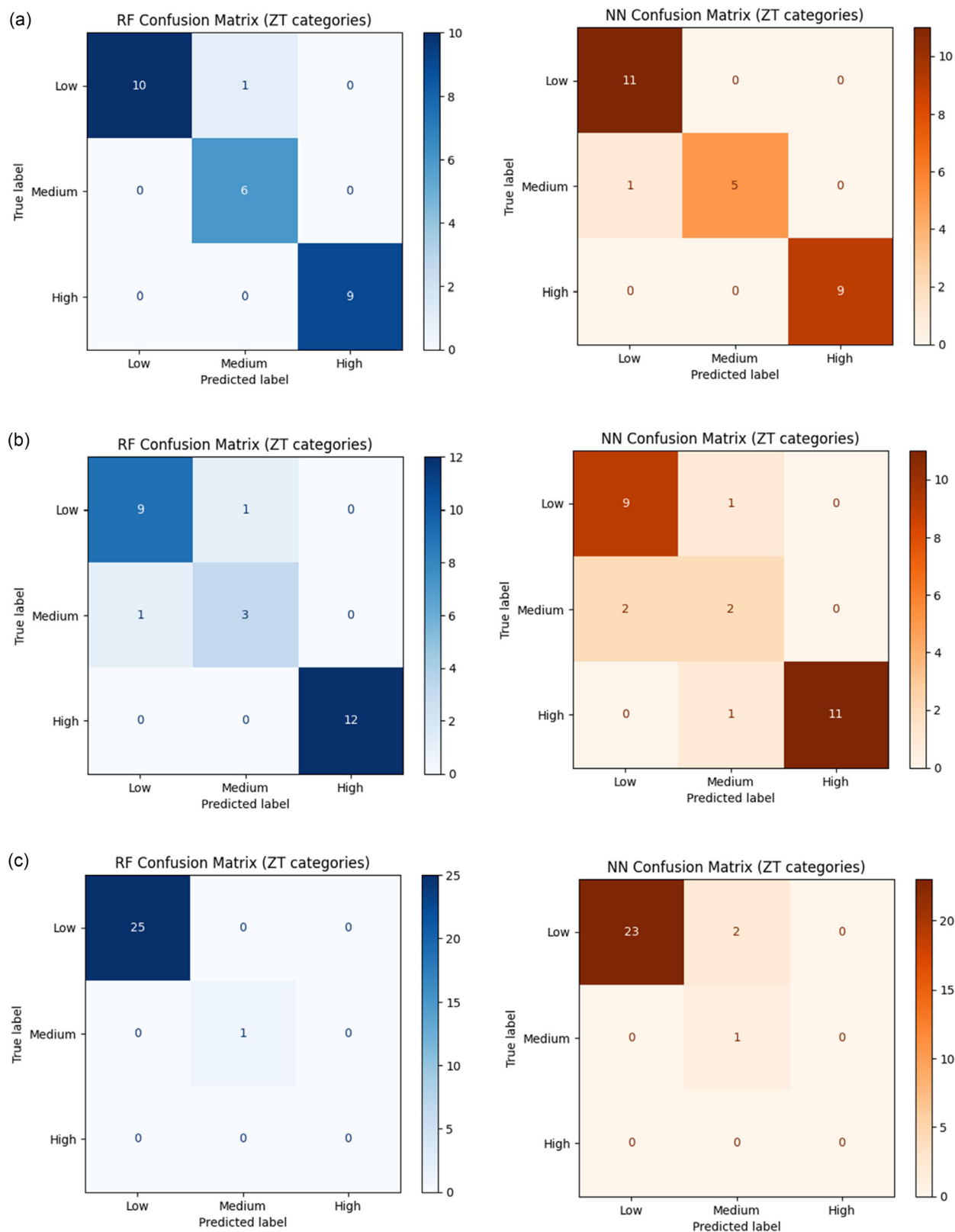


Figure 3. Confusion matrices for Random Forest (left) and Neural Network (right) classifiers at a) 100 K, b) 300 K, and c) 800 K, showing predicted versus actual ZT performance categories (Low: $ZT < 0.2$; Medium: $0.2 \leq ZT < 0.4$; High: $ZT \geq 0.4$).

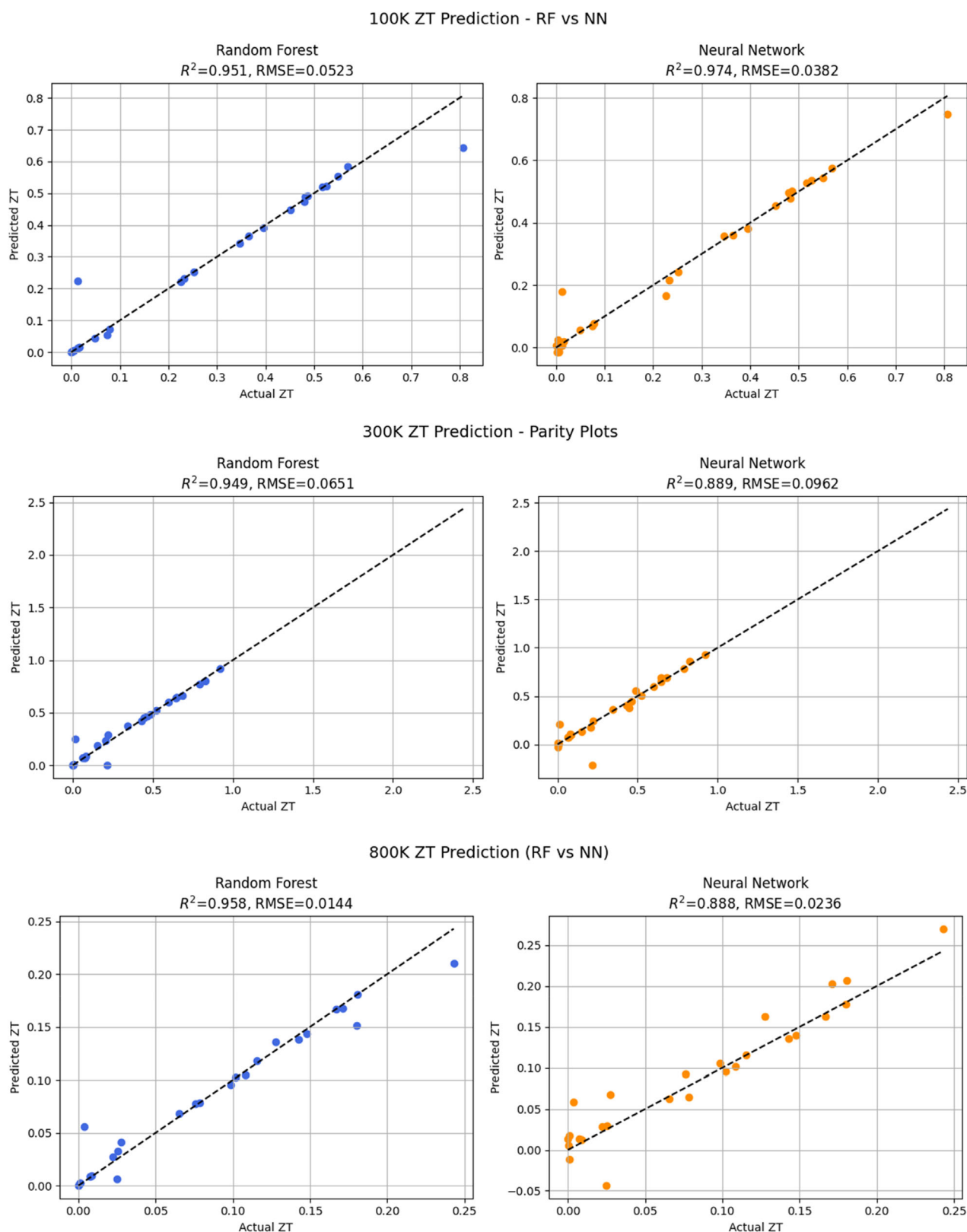


Figure 4. Parity plots comparing predicted versus actual ZT values using Random Forest (left) and Neural Network (right) models at 100 K (top row), 300 K (middle row), and 800 K (bottom row). The coefficient of determination (R^2) and RMSE are reported for each model. Both models demonstrate strong agreement with actual values across temperatures, with Neural Networks showing slightly better generalization at 100 K and 300 K, and Random Forests achieving superior performance at 800 K.

in transport properties probably helps predictions generalize better and have less variation. Both models still had very high predictive accuracy at 300 K. The RF model had an R^2 of 0.949 and an RMSE of 0.0651. The NN model had a slightly lower R^2 of 0.889 and a higher RMSE of 0.0962. These findings indicate that the RF model may more effectively represent the nonlinearity in thermoelectric behavior at room temperature, where augmented phonon–electron interactions start to affect transport properties in a more variable manner. The slightly worse performance of the NN could also be due to sensitivity to local minima during training or hyperparameter convergence that is not as good in this middle range. This observation is consistent with findings from Deng et al.,^[38] which indicated that RF models frequently surpass NN models in predicting thermoelectric properties due to their resilience in managing intricate feature interactions. A change in the relative performance of the models was seen at 800 K. The RF model once more beat the NN model, getting $R^2 = 0.958$ and a very low RMSE = 0.0144. The NN model, on the other hand, had $R^2 = 0.888$ and RMSE = 0.0236. The RF's better performance at high temperatures may be because it is more resistant to noise and overfitting when used in groups, especially when thermal excitation and lattice scattering make the underlying physics more variable. Also, because the range of ZT values gets smaller at 800 K, RF's ability to model nonlinear thresholds without extrapolation may make it more accurate than the NN's smooth function approximations. This is consistent with the study by Fan and Oganov,^[39] which demonstrated the effectiveness of ensemble models like RF in high-temperature thermoelectric predictions. From a physical perspective, the models' predictive success across all temperatures highlights the strong coupling between ZT and its governing variables: Seebeck coefficient, electrical conductivity, thermal conductivity, and chemical potential. The high correlation seen in all parity plots confirms that ML frameworks can accurately encode the complex interplay between these parameters, even as they evolve with temperature. Notably, the models demonstrate strong reliability

even under conditions of sparse data (e.g., at 800 K where high-ZT examples are rare), supporting their utility in materials screening and thermoelectric optimization. This underscores the potential of ML models in accelerating the discovery of high-performance thermoelectric materials, as also suggested by Kim et al.^[36] In addition to test set performance, we evaluated model accuracy on the training sets. The training metrics (R^2 and RMSE) are consistent with the test results, as summarized in Table S2 Supporting Information, indicating that both RF and NN models generalize well without significant overfitting or underfitting.

Figure 5 shows a comparison of the best thermoelectric materials at three different temperatures: 100 K (blue), 300 K (orange), and 800 K (green). Each marker shows the highest ZT value found for a certain compound using ML analysis at different chemical potentials. The figure shows how thermoelectric performance changes with temperature and points out the materials that always have a high ZT. It is clear that there are a few trends. First, ZT values are usually higher at 300 K than at 100 K or 800 K. This is because the increased carrier mobility and manageable thermal conductivity are in balance. At low temperatures, ZT values are more tightly grouped between 0.4 and 0.8. This means that performance does not vary as much because phonons are suppressed and intrinsic carrier behavior is more important. When the temperature is high (800 K), most materials show a big drop in ZT. This is because phonon scattering and lattice thermal conductivity both go up. Under these conditions, only a few compounds still work well.

Rb₂GeI₆ is the most promising of the compounds tested. It has a maximum ZT of 2.44 and an average ZT of 1.10 at all temperatures. Its high ZT at 300 K and stable performance at 100 K and 800 K show that it is strong and can handle different carrier concentrations and temperatures. Rb₂PbI₆, which has a high ZT and is stable across a range of temperatures; Cs₂SnBr₆, which does especially well at 100 K and 300 K; and In₂PtCl₆, which is a rare example of a heavy-metal halide with competitive ZT

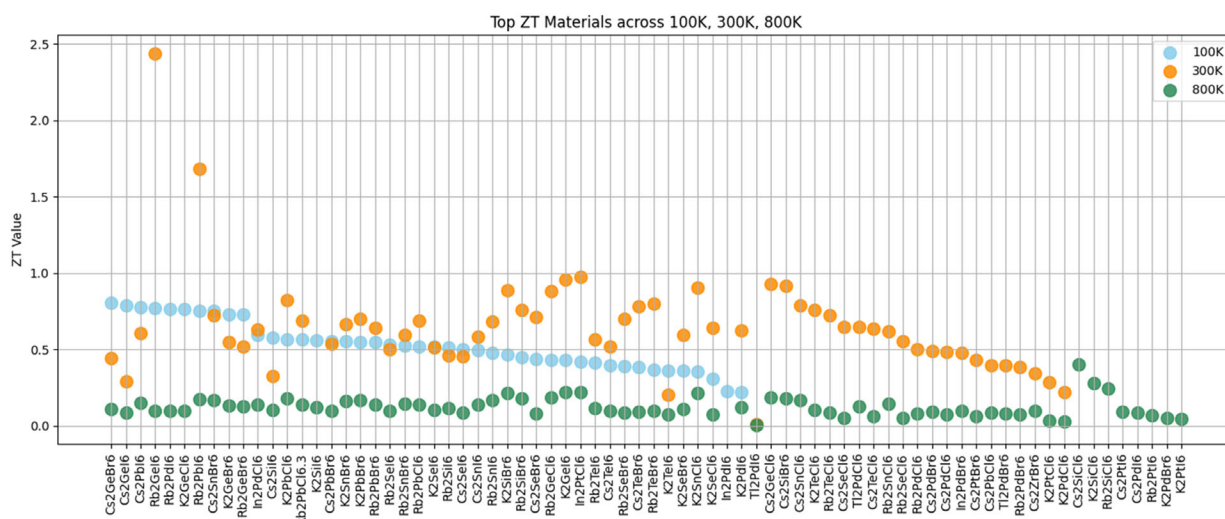


Figure 5. Peak thermoelectric figure of merit (ZT) for top-performing materials at 100 K (blue), 300 K (orange), and 800 K (green). Each point represents the maximum ZT achieved across the explored chemical potential range for a given material. The figure highlights compounds with strong and stable thermoelectric performance across multiple temperatures, including Rb₂GeI₆, Rb₂PbI₆, Cs₂SnBr₆, and In₂PtCl₆.

at mid-to-high temperatures. These findings underscore the capability of certain halide perovskite derivatives to function as tunable, thermally stable thermoelectric materials. Taken together, the figure and supporting analysis demonstrate the strength of the ML framework in not only identifying high-ZT compounds but also mapping their thermal performance landscape. This provides a critical tool for guiding material selection across practical operating regimes in thermoelectric applications.

3.3. Influence of Chemical Potential on Thermoelectric Performance

To explore more closely the impact of chemical potential (μ) on thermoelectric performance, we conducted a focused ML

analysis utilizing the 300 K dataset. The chemical potential (μ) was varied between -2 and $+2$ eV to provide a broad scan across both p -type and n -type regimes. While not all μ values in this range correspond to realistic carrier concentrations, this wide window ensures that all band-edge regions are included. In practice, only μ values close to the band edges (typically within ≈ 0.1 – 0.5 eV) correspond to experimentally relevant carrier concentrations ($\approx 10^{18}$ – 10^{21} cm $^{-3}$), and our interpretation of ZT trends is restricted to these physically meaningful regions. Two types of models were constructed: one set incorporating μ alongside the key transport descriptors and a second, simplified model using μ as the sole input feature. The comparative performance of these models is shown in **Figure 6**. The parity plot for the μ -only RF model is in panel (a), and the parity plots for the full-feature RF and NN models are in panels (b) and (c).

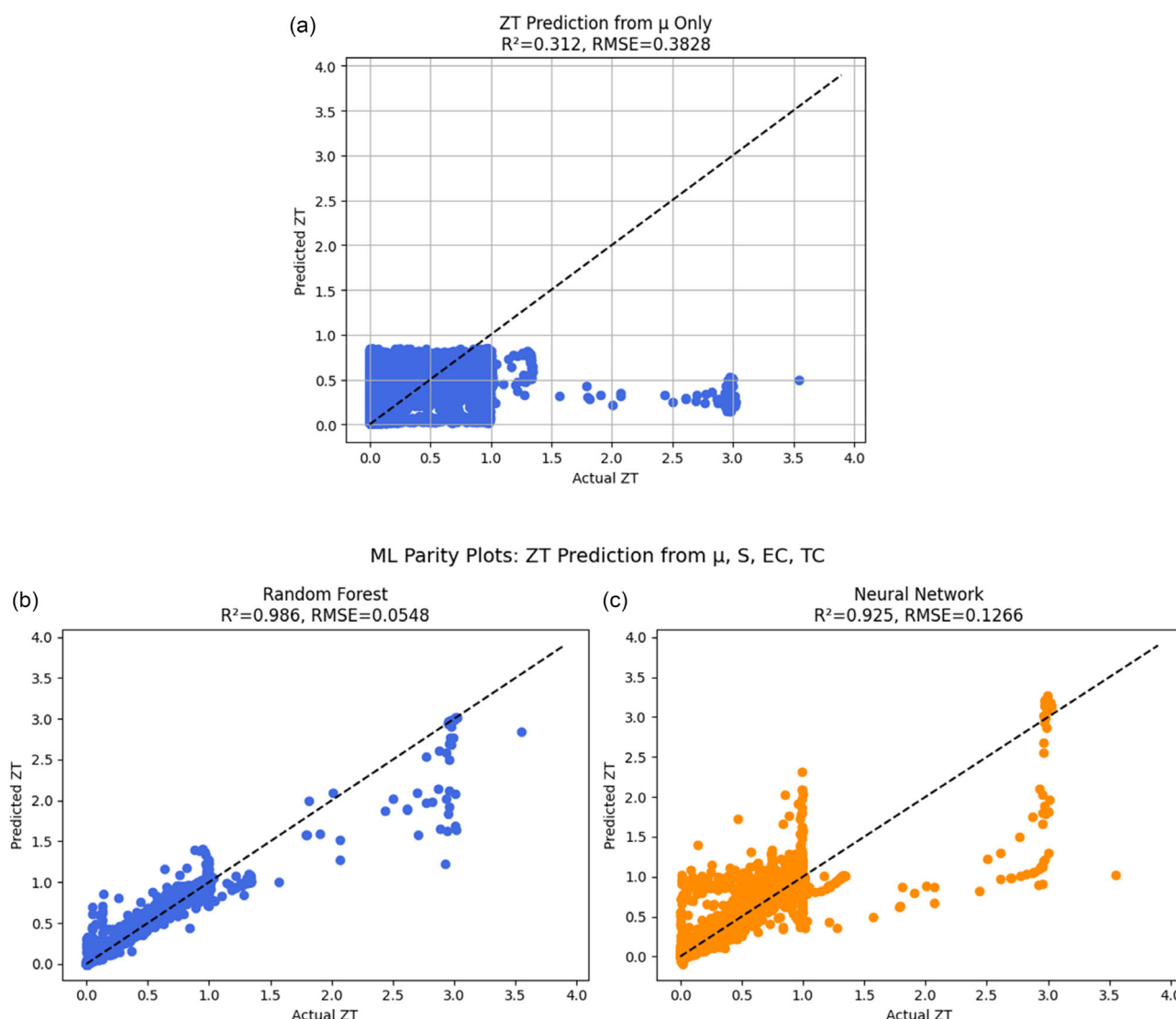


Figure 6. Parity plots comparing predicted versus actual ZT values at 300 K using different ML models. a) Prediction using chemical potential (μ) as the sole input feature. b) RF and c) NN models trained on the full feature set, including μ , Seebeck coefficient (S), electrical conductivity (σ), and thermal conductivity (κ). The coefficient of determination (R^2) and RMSE are reported for each model. The figure highlights the essential yet insufficient predictive role of μ in isolation, and the benefit of including additional transport features.

The full-feature models show that they can make very accurate predictions. The RF model had an R^2 value of 0.986 and an RMSE value of 0.0548, which means that the predicted and actual ZT values were almost exactly the same. The NN model was very close behind, with an R^2 of 0.925 and an RMSE of 0.1266. These results show that the ML models can learn complicated, nonlinear relationships between the input features and make very accurate predictions when they have all the transportation data. The close clustering of predicted points along the identity line in both parity plots underscores the reliability of this ML framework for high-throughput thermoelectric screening.

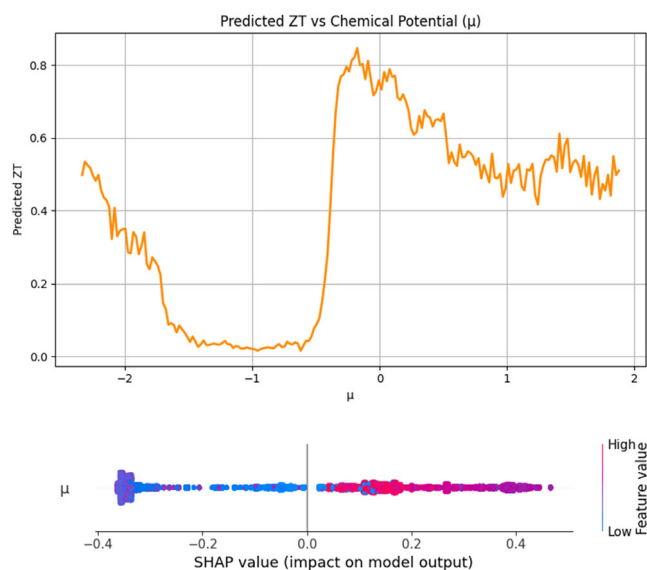


Figure 7. Predicted ZT variation as a function of chemical potential (μ) at 300 K (top panel), alongside SHAP summary plot showing the influence of μ on the ML model's ZT prediction (bottom panel). The upper curve highlights the strong, non-monotonic dependence of ZT on μ , with optimal performance observed near $\mu \approx -0.1$ eV. The SHAP beeswarm plot quantifies μ 's contribution, indicating that high μ values systematically enhance the predicted ZT.

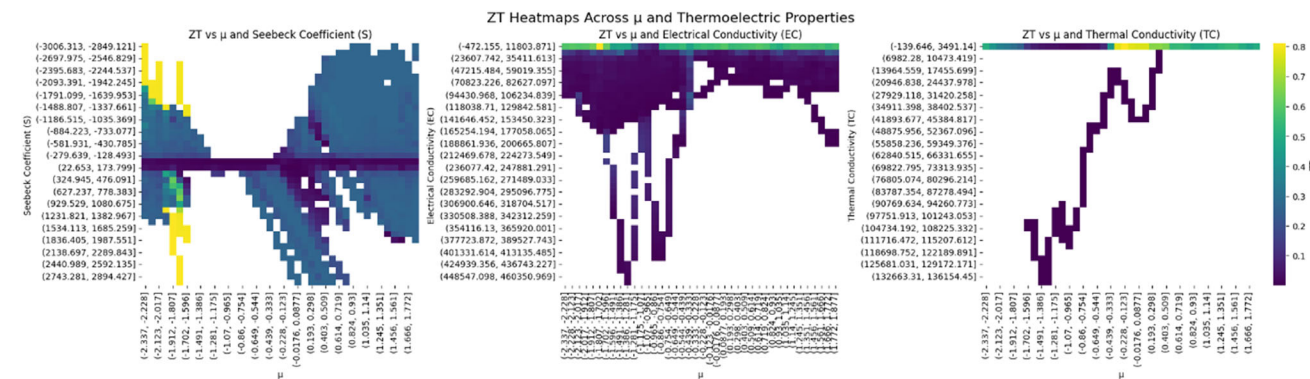


Figure 8. Heatmaps illustrating the predicted thermoelectric figure of merit (ZT) at 300 K as a function of chemical potential (μ) and key transport properties. (Left) ZT versus μ and Seebeck coefficient (S), showing a broad optimal ZT plateau at intermediate μ and moderate S values. (Center) ZT versus μ and electrical conductivity (EC), highlighting the synergistic balance between EC and μ in maximizing ZT, with saturation observed at high EC. (Right) ZT versus μ and thermal conductivity (TC), demonstrating the expected preference for low TC regions but with weaker μ sensitivity. The plots confirm that ZT optimization requires tuning μ within a specific window ($\mu \approx -0.3$ eV to $+0.5$ eV) to achieve a favorable trade-off between S, EC, and TC.

To isolate the individual impact of μ , we trained an RF model using μ alone. As shown in Figure 6a, this model yields a markedly lower $R^2 = 0.312$ and a higher $\text{RMSE} = 0.3828$. This decline in predictive performance confirms that although μ exerts a measurable influence on ZT, it cannot independently account for thermoelectric behavior. This is consistent with the role of μ in determining carrier concentration and Fermi-level alignment, which are necessary but not sufficient for high ZT. The overall thermoelectric efficiency also depends on how these carriers interact with the material's band structure, phonon spectrum, and scattering mechanisms. S, σ , and κ are better at showing these properties.

Nevertheless, the fact that μ alone accounts for more than 30% of the variance in ZT is important. It emphasizes μ as a crucial thermodynamic parameter for optimizing performance, especially via carrier engineering techniques like doping or band structure modulation. SHAP interpretability analysis backs this up even more by consistently placing μ among the most important factors in high-ZT predictions. Therefore, μ can be considered both a rapid pre-screening metric and a physical tuning parameter, especially useful in guiding material synthesis or electronic structure optimization.

Figure 7 shows how chemical potential (μ) directly affects thermoelectric performance, as shown by the RF model at 300 K. The upper panel shows the predicted ZT based on μ , and the lower panel shows the SHAP summary plot, which shows how much the feature added to the model output. The predicted ZT shows a very non-monotonic pattern in relation to μ . Between $\mu \approx -2.0$ eV and $\mu \approx -1.0$ eV, ZT drops quickly, hitting a clear low point at $\mu \approx -1.0$ eV. This behavior shows that the Fermi level is deep within the bandgap, which stops electronic transport and lowers ZT. When μ gets close to -0.1 eV, ZT reaches a maximum of about 0.8. This is because the Fermi level lines up well with the edges of the conduction or valence bands, which leads to the best carrier concentration and a higher power factor. After $\mu = 0.5$ eV, ZT starts to go down again, probably because bipolar conduction increases and parasitic transport effects start occurring. The SHAP beeswarm plot in the lower panel backs up these findings by showing how much μ adds to the predicted ZT across the

whole dataset. High μ values (in pink) are always linked to positive SHAP values, which means that raising μ helps ZT prediction. On the other hand, low μ values (blue) can harm ZT. The distribution of SHAP values also shows that μ alone cannot fully predict ZT (see Figure 7a), but it still has a big effect, especially in the areas where the best carrier concentration can be reached. From a physical standpoint, these findings illustrate the dual function of μ in regulating both carrier density and the alignment of the Fermi level in relation to the transport-relevant bands. One can move the system toward areas of higher ZT by changing μ through doping or alloying. The wide μ window for high ZT (from about -0.3 eV to 0.5 eV) indicates that the material family being studied can handle changes in carrier concentration, which is advantageous to creating devices that perform well.

To better understand how the chemical potential affects thermoelectric performance by changing important transport properties, we made 2D heatmaps of predicted ZT as a function of μ and each important descriptor: Seebeck coefficient (S), electrical conductivity (named EC and previously σ), and thermal conductivity (TC also symbolized κ in the previous sections); at 300 K. Figure 8 shows the results in a way that fully shows the μ -dependent transport landscape. The left side of Figure 8 shows that ZT has a strong and very nonlinear relationship with both μ and S . When μ is low ($\mu < -1$ eV), ZT is usually low as well. This is because the Fermi level is deep within the bandgap, which means that there are few carriers and poor thermoelectric activity. As μ gets closer to the edges

of the conduction or valence bands ($\mu \approx -0.1$ eV to $+0.5$ eV), the Seebeck coefficient stays high enough, and the carrier concentration goes up, creating a wide plateau of higher ZT. Interestingly, areas with very high S , especially those over $-2000 \mu\text{V K}^{-1}$, have low ZT. This shows the trade-off between S and σ that is often seen in semiconductors because of the Mott relation, which shows the opposite relationship. The middle panel, which shows ZT against μ and EC, shows how important it is to find the right balance between electrical transport and μ . Increasing EC usually makes ZT better at the right μ , but after a certain point, the benefit levels off. This is probably because S goes down at the same time, and bipolar conduction effects start to show up. The best ZT regions seem to be in a certain μ range ($\mu = -0.3$ eV to 0.5 eV) and moderate EC values. This shows that ZT is not just maximized by maximizing EC, but also by adjusting μ to find the right balance between EC and S . The right panel depicts the effect of μ on ZT through its relationship with TC. As expected, the dependency is weaker compared to S and EC, since TC is predominantly phonon-limited and less sensitive to electronic chemical potential at fixed lattice structures. Nonetheless, regions of higher ZT correspond to lower TC, particularly when μ aligns favorably with the conduction or valence band edges. This suggests that μ tuning alone cannot sufficiently reduce TC and that combining μ engineering with strategies to suppress lattice thermal conductivity (e.g., alloying, nanostructuring) remains essential for achieving superior thermoelectric performance. Taken together, these heatmaps

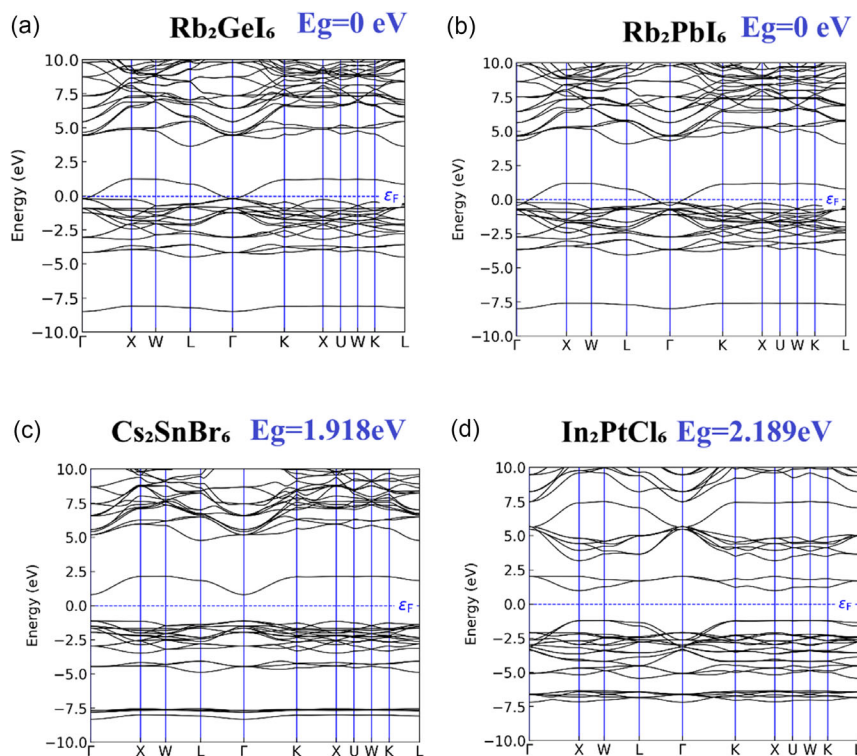


Figure 9. Electronic band structures of a) Rb_2GeI_6 , b) Rb_2PbI_6 , c) Cs_2SnBr_6 , and d) In_2PtCl_6 , showing band dispersion along high-symmetry k -points and indicating direct or indirect bandgap character.

underscore that the influence of μ on ZT is not isolated but strongly mediated by its coupled effects on S, EC, and, to a lesser extent, TC. The results validate that optimal ZT is achieved within a specific μ range (≈ -0.3 eV to $+0.5$ eV), where favorable trade-offs between high S, sufficient EC,

and low TC are realized. This reinforces the importance of μ as a tuning parameter within a multivariate design space, providing quantitative guidance for carrier concentration optimization alongside material engineering strategies targeting lattice and band structure.

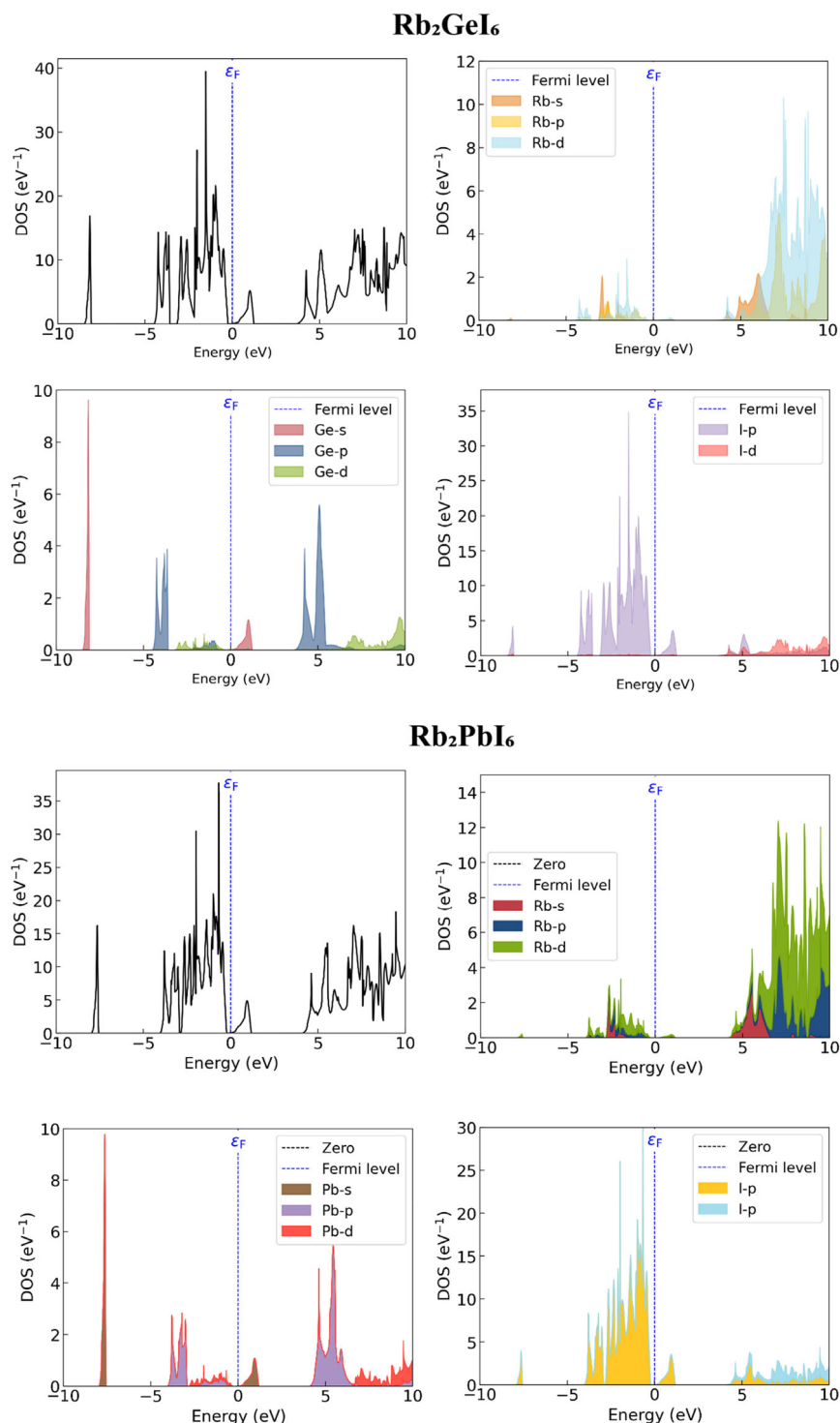


Figure 10. Total and partial density of states (TDOS/PDOS) for each compound, decomposed into atomic orbital contributions (*s*, *p*, *d* states), highlighting the hybridization near the Fermi level.

In₂PtCl₆

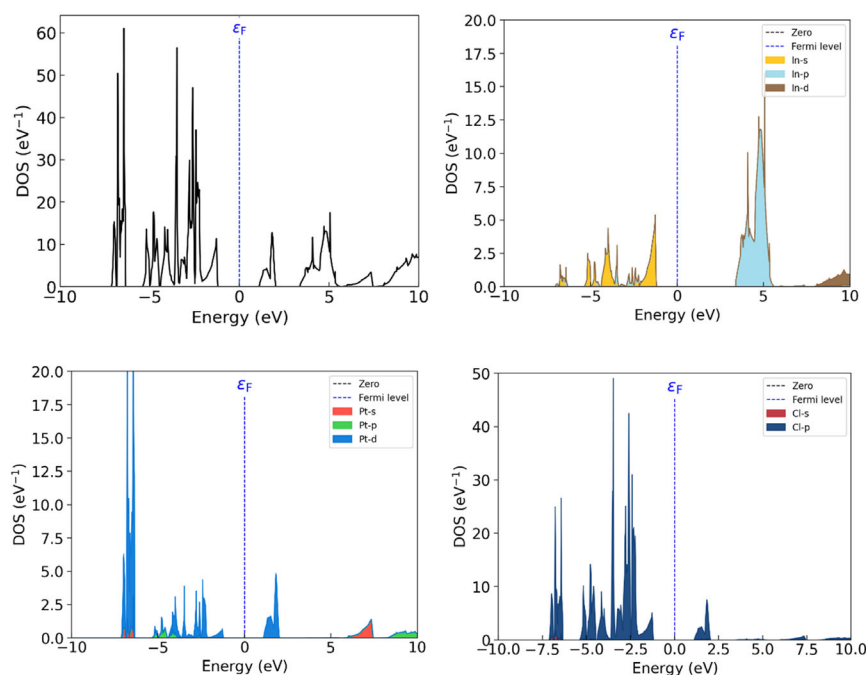


Figure 10. Continued.

3.4. Electronic Structure and its Relationship to Thermoelectric Performance

We looked at the electronic band structures and DOS of the top-performing compounds found in this study: Rb₂GeI₆, Rb₂PbI₆, Cs₂SnBr₆, and In₂PtCl₆, to help us understand their thermoelectric behavior. **Figures 9** and **Figure 10** show what we found. The compounds Rb₂GeI₆ and Rb₂PbI₆ exhibit metallic or degenerate semiconducting characteristics, with their Fermi levels intersecting partially filled conduction bands ($E_g \approx 0$ eV). This electronic configuration makes the material very conductive, which directly improves the power factor ($S^2\sigma$). However, the Seebeck coefficient tends to be lower in such metallic systems due to the high carrier concentration and the flatness of the DOS near E_F . Nevertheless, the observed hybridization between Rb-*d* and I-*p*/Ge-*p*/Pb-*p* orbitals contributes to improved carrier mobility, supporting their moderate thermoelectric performance.

On the other hand, Cs₂SnBr₆ and In₂PtCl₆ are indirect bandgap semiconductors, with E_g values of 1.918 eV and 2.189 eV, respectively. Their electronic structures have conduction and valence bands that are well-separated. The conduction band is made up of Sn-*p*/Br-*p* orbitals, and the valence band is made up of Pt-*d*/Cl-*p* orbitals. These wide bandgaps stop intrinsic carrier excitation at low temperatures, which helps keep the Seebeck coefficient high. Also, the sharp DOS features near the band edges make thermopower stronger by making the distribution of carriers around E_F more uneven. The difference in bandgap character also influences the ZT values observed. The metallic compounds (Rb₂GeI₆, Rb₂PbI₆) benefit from high σ but are limited by lower S and increased electronic thermal conductivity (κ_e). Conversely, the

semiconducting systems (Cs₂SnBr₆, In₂PtCl₆) achieve better thermoelectric balance due to their high S and reduced κ_e , even though their σ is comparatively lower. This trade-off shows how important it is to optimize both carrier concentration and band structure at the same time for thermoelectric materials that work well. In general, the analysis of the electronic structure shows that both metallic and semiconducting behavior can help thermoelectric efficiency, depending on how the electrical and thermal transport properties are balanced. The results also show that halide-based perovskite-like materials can be made in many different ways, which supports their potential for use in thermoelectric applications.

4. Conclusion

In this study, we conducted a comprehensive theoretical investigation of 102 halide double perovskites using a hybrid DFT–Boltzmann transport framework complemented by ML analysis. Hybrid-functional (HSE06) band structures provided accurate electronic descriptors, which were used to compute temperature-dependent thermoelectric transport coefficients via Boltzmann theory. This led to the identification of several promising compounds, including Rb₂GeI₆, Rb₂PbI₆, Cs₂SnBr₆, and In₂PtCl₆, exhibiting favorable Seebeck coefficients, electrical conductivities, and figure of merit (ZT) values. Our ML models, which used both RF and NN, made very accurate predictions of ZT across different temperature ranges. The RF model was very easy to understand and worked well, especially for general screening and mid-range temperature applications. The NN model, on the other hand, was better at generalizing to capture subtle, high-ZT outliers, especially at high temperatures. These insights highlight the significance of selecting

models that correspond with dataset balance, application domain, and operational conditions. ML also made it possible to use SHAP analysis to figure out which features are important, which confirmed that the chemical potential (μ) is the main factor that affects thermoelectric performance. This confirms a two-step discovery process: first, μ -based screening to narrow down the search space, and then full-feature ML or DFT modeling of the best candidates. This strategy not only makes predictions more accurate, but it also gives guidelines for how to dope and tune carrier concentration. Our results are consistent with the increasing evidence that machine learning has emerged as a prominent and influential instrument in thermoelectric materials research. In the last ten years, ML algorithms, such as support vector machines and deep neural networks, have been able to predict Seebeck coefficients, electrical conductivity, and ZT for a wide range of materials, such as oxides, chalcogenides, skutterudites, and half-Heuslers. The synergy between data-driven and physics-based approaches will be crucial for speeding up the discovery of next-generation thermoelectric materials as larger and better-quality datasets become available and machine learning architectures like graph neural networks, attention mechanisms, and generative models improve. In conclusion, our combination of precise electronic structure calculations with Boltzmann transport modeling and interpretable machine learning offers a scalable method to predict, comprehend, and design high-ZT halide perovskites. This hybrid framework has a lot of potential for making clean, efficient energy conversion materials that are tailored to specific needs.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

Boltzmann transport theory, halide double perovskites, hybrid density functional theory (HSE06), machine learning screening, thermoelectric materials

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